An Analysis of Variational autoencoder

Variational autoencoder

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$$

Variational autoencoder -- Part 1

To efficiently search for θ ∗ , φ ∗ = argmax θ , φ L θ , φ (x) {\displaystyle \theta ^{*},\phi ^{*}={\underset {\theta ,\phi }{\operatorname {argmax} }}\,L_{\theta ,\phi }(x)} the typical method is gradient ascent. It is straightforward to find ∇ θ E z ∼ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] = E z ∼ q φ (· | x) [∇ θ ln ■ p θ (x , z) q φ (z | x)] {\displaystyle \nabla _{\theta }\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\nabla _{\theta }\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]} However, ∇ φ E z ∼ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] {\displaystyle \nabla _{\phi }\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]} does not allow one to put the ∇ φ {\displaystyle \nabla _{\phi }} inside the expectation, since φ {\displaystyle \phi } appears in the probability distribution itself. The reparameterization trick (also known as stochastic backpropagation) bypasses this difficulty. The most important example is when z ∼ q φ (· | x) {\displaystyle z\sim q_{\phi }(\cdot |x)} is normally distributed, as N (μ φ (x) , Σ φ (x)) {\displaystyle {\mathcal {N}}(\mu _{\phi }(x),\Sigma _{\phi }(x))} .

Variational autoencoder -- Part 2

the sliced Wasserstein distance used by S Kolouri, et al. in their VAE the energy distance implemented in the Radon Sobolev Variational Auto-Encoder the Maximum Mean Discrepancy distance used in the MMD-VAE the Wasserstein distance used in the WAEs kernel-based distances used in the Kernelized Variational Autoencoder (K-VAE)

Variational autoencoder -- Part 3

where p θ (x , z) {\displaystyle p_{\theta }({x,z})} represents the joint distribution under p θ {\displaystyle p_{\theta }} of the observable data x {\displaystyle x} and its latent representation or encoding z {\displaystyle z} . According to the chain rule, the equation can be rewritten as

Variational autoencoder -- Part 4

the usual reconstruction error part which seeks to ensure that the encoder-then-decoder mapping x ■ D θ (E ψ (x)) {\displaystyle x\mapsto D_{\theta }(E_{\psi }(x))} is as close to the identity map as possible; the sampling is done at run time from the empirical distribution P r e a l {\displaystyle \mathbb {P} ^{real}} of objects available (e.g., for MNIST or IMAGENET this will be the empirical probability law of all images in the dataset). This gives the term: E x ∼ P r e a l [■ x − D θ (E φ (x)) ■ 2 2] {\displaystyle \mathbb {E} _{x\sim \mathbb {P} ^{real}}\left[\|x-D_{\theta }(E_{\phi }(x))\|_{2}^{2}\right]} . a variational part that ensures that, when the empirical distribution P r e a l {\displaystyle \mathbb {P} ^{real}} is passed through the encoder E φ {\displaystyle E_{\phi }} , we recover the target distribution, denoted here μ (d z) {\displaystyle \mu (dz)} that is usually taken to be a Multivariate normal distribution. We will denote E φ ■ P r e a l {\displaystyle E_{\phi }\sharp \mathbb {P} ^{real}} this pushforward measure which in practice is just the empirical distribution obtained by passing all dataset objects through the encoder E φ {\displaystyle E_{\phi }} . In order to make sure that E φ ■ P r e a l {\displaystyle E_{\phi }\sharp \mathbb {P} ^{real}} is close to the target μ (d z) {\displaystyle \mu (dz)} , a Statistical distance d {\displaystyle d} is invoked and the term d (μ (d z) , E φ ■ P r e a l) 2 {\displaystyle d\left(\mu (dz),E_{\phi }\sharp \mathbb {P} ^{real}\right)^{2}} is added to the loss. We obtain the final formula for the loss:

Variational autoencoder -- Part 5

The statistical distance d {\displaystyle d} requires special properties, for instance it has to be posses a formula as expectation because the loss function will need to be optimized by stochastic optimization algorithms. Several distances can be chosen and this gave rise to several flavors of VAEs:

Variational autoencoder -- Part 6

D K L (q φ (z | x) ■ p θ (z | x)) = E z ∼ q φ (· | x) [ln ■ q φ (z | x) p θ (z | x)] = E z ∼ q φ (· | x) [ln ■ q φ (z | x) p θ (x) p θ (x , z)] = ln ■ p θ (x) + E z ∼ q φ (· | x) [ln ■ q φ (z | x) p θ (x , z)] . {\displaystyle {\begin{aligned}D_{KL}(q_{\phi }({z|x})\parallel p_{\theta }({z|x}))&=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }(z|x)}{p_{\theta }(z|x)}}\right]\\&=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }({z|x})p_{\theta }(x)}{p_{\theta }(x,z)}}\right]\\&=\ln p_{\theta }(x)+\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }({z|x})}{p_{\theta }(x,z)}}\right].\end{aligned}}}

Variational autoencoder -- Part 7

Now, define the evidence lower bound (ELBO): L θ , φ (x) := E z ∼ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] = ln ■ p θ (x) − D K L (q φ (· | x) ■ p θ (· | x)) {\displaystyle L_{\theta ,\phi

}(x):=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]=\ln p_{\theta }(x)-D_{KL}(q_{\phi }({\cdot |x})\parallel p_{\theta }({\cdot |x}))} Maximizing the ELBO θ ∗ , φ ∗ = argmax θ , φ L θ , φ (x) {\displaystyle \theta ^{*},\phi ^{*}={\underset {\theta ,\phi }{\operatorname {argmax} }}\,L_{\theta ,\phi }(x)} is equivalent to simultaneously maximizing ln ■ p θ (x) {\displaystyle \ln p_{\theta }(x)} and minimizing D K L (q φ (z | x) ■ p θ (z | x)) {\displaystyle D_{KL}(q_{\phi }({z|x})\parallel p_{\theta }({z|x}))} . That is, maximizing the log-likelihood of the observed data, and minimizing the divergence from the approximate posterior q φ (· | x) {\displaystyle q_{\phi }(\cdot |x)} to the exact posterior p θ (· | x) {\displaystyle p_{\theta }(\cdot |x)} . The form given is not very convenient for maximization, but the following, equivalent form, is: L θ , φ (x) = E z ~ q φ (· | x) [ln ■ p θ (x | z)] − D K L (q φ (· | x) ■ p θ (·)) {\displaystyle L_{\theta ,\phi }(x)=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln p_{\theta }(x|z)\right]-D_{KL}(q_{\phi }({\cdot |x})\parallel p_{\theta }(\cdot))} where ln ■ p θ (x | z) {\displaystyle \ln p_{\theta }(x|z)} is implemented as − 1 2 ■ x − D θ (z) ■ 2 2 {\displaystyle -{\frac {1}{2}}\|x-D_{\theta }(z)\|_{2}^{2}} , since that is, up to an additive constant, what x | z ~ N (D θ (z) , I) {\displaystyle x|z\sim {\mathcal {N}}(D_{\theta }(z),I)} yields. That is, we model the distribution of x {\displaystyle x} conditional on z {\displaystyle z} to be a Gaussian distribution centered on D θ (z) {\displaystyle D_{\theta }(z)} . The distribution of q φ (z | x) {\displaystyle q_{\phi }(z|x)} and p θ (z) {\displaystyle p_{\theta }(z)} are often also chosen to be Gaussians as z | x ~ N (E φ (x) , σ φ (x) 2 I) {\displaystyle z|x\sim {\mathcal {N}}(E_{\phi }(x),\sigma _{\phi }(x)^{2}I)} and z ~ N (0 , I) {\displaystyle z\sim {\mathcal {N}}(0,I)} , with which we obtain by the formula for KL divergence of Gaussians: L θ , φ (x) = − 1 2 E z ~ q φ (· | x) [■ x − D θ (z) ■ 2 2] − 1 2 (N σ φ (x) 2 + ■ E φ (x) ■ 2 2 − 2 N ln ■ σ φ (x)) + C o n s t {\displaystyle L_{\theta ,\phi }(x)=-{\frac {1}{2}}\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\|x-D_{\theta }(z)\|_{2}^{2}\right]-{\frac {1}{2}}\left(N\sigma _{\phi }(x)^{2}+\|E_{\phi }(x)\|_{2}^{2}-2N\ln \sigma _{\phi }(x)\right)+Const} Here N {\displaystyle N} is the dimension of z {\displaystyle z} . For a more detailed derivation and more interpretations of ELBO and its maximization, see its main page.